

Project 2: Numerical Solutions to Incompressible Navier–Stokes Equations

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The 2-D lid-driven cavity flow is solved using a fractional-step projection method on a staggered MAC grid. The incompressible Navier–Stokes equations are time-integrated with second-order Adams–Bashforth advection and Crank–Nicolson diffusion; a pressure Poisson equation is solved at each step to enforce the divergence-free constraint. Simulations are performed on a 65×65 uniform grid with $CFL = 0.1$ for Reynolds numbers $Re = 100, 400$, and 1000 . Computed centerline velocity profiles are compared against the benchmark data of Ghia et al., showing close agreement across all three Reynolds numbers. The fractional-step approximation yields first-order temporal accuracy while the staggered-grid spatial operators are second-order accurate, consistent with the theoretical properties of the scheme. Flow structures including the primary recirculation vortex and secondary corner vortices are well captured at all Reynolds numbers investigated.

I. INTRODUCTION

The lid-driven cavity flow problem is a computational fluid dynamics (CFD) benchmark problem that consists of simulating the flow of fluid within a square domain with three rigid walls and a driving velocity on the top boundary. The top lid moves at a constant velocity, while the other three walls have a no-slip boundary condition. This setup creates a complex flow pattern with a primary vortex in the center and secondary vortices in the corners, which evolve as the Reynolds number increases. The problem is widely used to validate numerical methods for incompressible flows due to its well-documented solutions and sensitivity to numerical accuracy. The fractional stepping method used in this project allows for separation of the velocity and pressure fields, which is crucial for solving the incompressible Navier–Stokes equations efficiently.

The objective of this project is to implement a fractional-step solver for the 2-D incompressible Navier–Stokes equations on a staggered MAC grid and to validate the results against the benchmark data of Ghia et al. for Reynolds numbers $Re = 100, 400$, and 1000 . The solver employs second-order Adams–Bashforth advection and Crank–Nicolson diffusion, with a conjugate gradient pressure Poisson solve at each time step to maintain the divergence-free condition.

II. GOVERNING EQUATIONS

The dimensionless 2-D incompressible Navier–Stokes equations are

$$\frac{\partial \mathbf{q}}{\partial t} + \mathbf{q} \cdot \nabla \mathbf{q} = -\nabla p + \frac{1}{Re} \nabla^2 \mathbf{q}, \quad (1)$$

$$\nabla \cdot \mathbf{q} = 0, \quad (2)$$

where $\mathbf{q} = \{u, v\}$ is the velocity field, p is pressure, and Re is the Reynolds number. The boundary conditions

are no-slip ($\mathbf{q} = \mathbf{0}$) on the bottom, left, and right walls, and a driven lid with $u = 1, v = 0$ on the top boundary ($y = 1$).

III. OPERATORS

III.A. Gradient Operator

The gradient operator ($\mathbf{G}$) is discretized using central differences for the interior points and one-sided differences for the boundary points. It maps pressure difference to velocity face values using the following formulas:

$$(\mathbf{G}p)_{i,j}^u = \frac{p_{i,j} - p_{i-1,j}}{\Delta x}, \quad (\mathbf{G}p)_{i,j}^v = \frac{p_{i,j} - p_{i,j-1}}{\Delta y} \quad (3)$$

In this case, the gradient operator is second order accurate in the interior.

III.B. Divergence Operators

The divergence operator ($\mathbf{D}$) is discretized using central differences for the interior points and one-sided differences for the boundary points. It maps velocity face values to cell-centered divergence. The central points are calculated using the following formula:

$$(\mathbf{D}\mathbf{u})_{i,j} = \frac{u_{i+1,j} - u_{i,j}}{\Delta x} + \frac{v_{i,j+1} - v_{i,j}}{\Delta y} \quad (4)$$

The full divergence operator is split into the sum of the interior and boundary contributions. The boundary contribution $\mathbf{bc}_D$ corrects the divergence stencil at cells adjacent to each wall: it subtracts the prescribed normal face velocity divided by the grid spacing from cells on the inflow side and adds it on the outflow side. For the no-slip walls ($u = v = 0$) the contribution vanishes; only the driven lid ($u = 1$ at $y = 1$) produces a non-zero $\mathbf{bc}_D$ along the top row of pressure cells.

III.C. Laplacian Operators

For the interior points:

$$(\mathbf{L}\mathbf{u})_{i,j} = \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{\Delta x^2} + \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{\Delta y^2} \quad (5)$$

III.D. Advection Operator

The advection operator $\mathbf{N}(\mathbf{q})$ evaluates the nonlinear convective term $\mathbf{q} \cdot \nabla \mathbf{q}$ in conservative form. For the u -component, fluxes at each control-volume face are formed by averaging adjacent velocity nodes:

$$(\mathbf{N}(\mathbf{q}))_{i,j}^u = \frac{(u_{i,j}^R)^2 - (u_{i,j}^L)^2}{\Delta x} + \frac{u_{i,j}^T v_{i,j}^T - u_{i,j}^B v_{i,j}^B}{\Delta y}, \quad (6)$$

where superscripts R, L, T, B denote right, left, top, and bottom face-averaged values. An analogous expression is used for the v -component. The central face averaging renders the operator second-order accurate in space, and the conservative form ensures that the global kinetic-energy production by advection is $O(\Delta x^2)$, as verified by unit test A2.

III.E. Unit Testing of Operators

Table 1 summarizes the unit tests performed for each discrete operator, verifying their correctness and expected convergence properties. All tests passed successfully, confirming the accuracy of the implemented operators. With these results in mind, we can now move on to using the solver on the lid-driven cavity flow problem.

IV. NUMERICAL METHOD

IV.A. Staggered Grid Formulation

The staggered grid arrangement is illustrated in Figure 1, where the horizontal velocity u is defined at the vertical cell faces, the vertical velocity v is defined at the horizontal cell faces, and the pressure p is defined at the cell centers. This helps to avoid incorrect pressure calculations and ensures a consistent discretization of the divergence and gradient operators. The pressure at the bottom-left cell is pinned to zero to provide a reference for the pressure field.

IV.B. Spatial Discretization

The discrete operators $\mathbf{G}$, $\mathbf{D}$, $\mathbf{L}$, and $\mathbf{N}$ are assembled as sparse matrices acting on flattened 1-D velocity and pressure vectors in row-major order. The gradient $\mathbf{G}$ and divergence $\mathbf{D}$ use forward and backward differences at individual staggered faces; although each is formally first-order at a single face, their composition $\mathbf{D}\mathbf{G}$ approximates the scalar Laplacian to second order on the staggered grid. The Laplacian $\mathbf{L}$ uses standard second-order central differences. Boundary contributions to $\mathbf{D}$

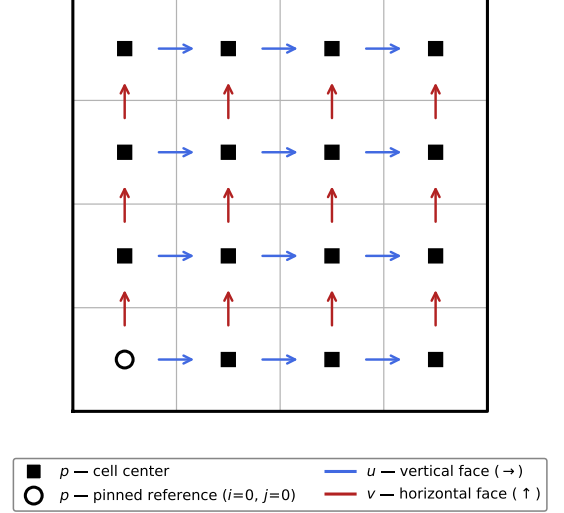


FIG. 1 Staggered grid formulation. Horizontal velocity u ($\rightarrow$), vertical velocity v ($\uparrow$), and pressure p ($\blacksquare$) are located at cell faces and cell centers, respectively. The open circle ($\circ$) marks the pinned pressure reference.

and $\mathbf{L}$ are collected in the correction vectors bc_D and bc_L and added to the right-hand side at each time step.

IV.C. Time Integration: Fractional-Step Method

Using second-order Adams–Bashforth for advection and Crank–Nicolson for diffusion, the semi-discrete momentum equation and continuity constraint form the saddle-point system.

$$\begin{pmatrix} A & G \\ D & 0 \end{pmatrix} \begin{pmatrix} \mathbf{q}^{n+1} \\ p^{n+1} \end{pmatrix} = \begin{pmatrix} \mathbf{r}^n + \text{bc}_1 \\ 0 \end{pmatrix}, \quad (7)$$

where $A = I/\Delta t - L/(2Re)$ and $\mathbf{r}^n$ contains the explicit Adams–Bashforth advection and Crank–Nicolson diffusion terms. Applying block LU decomposition and approximating $A^{-1} \approx \Delta t I$ yields the three-step fractional-step algorithm:

$$A\mathbf{q}^* = \mathbf{r}^n + \text{bc}_1, \quad (8)$$

$$D \Delta t G p^{n+1} = D\mathbf{q}^*, \quad (9)$$

$$\mathbf{q}^{n+1} = \mathbf{q}^* - \Delta t G p^{n+1}. \quad (10)$$

The intermediate velocity system (8) and pressure Poisson equation (9) are solved at each time step using a conjugate gradient method.

IV.D. Time Step and Stopping Criterion

The time step is chosen as

$$\Delta t = \min \left(\text{CFL} \frac{\Delta x}{u_{\text{top}}}, \text{CFL} (\Delta x)^2 Re \right), \quad (11)$$

and time integration proceeds until the steady-state criterion $\max |\mathbf{q}^{n+1} - \mathbf{q}^n| < 10^{-8}$ is satisfied.

TABLE I Unit tests for discrete operators, verifying correctness of each component.

Test	Operator	What is checked	Expected result	Pass?
G1	G	Gradient of linear p	Exact (machine ε)	✓
G2	G	Gradient of quadratic p	Error $O(\Delta x^2)$	✓
D1	DG	Composition $\approx \nabla^2$	Error $O(\Delta x^2)$ interior	✓
D2	D	Divergence of div-free field	Error $O(\Delta x^2)$ interior	✓
D3	D, G	Adjoint: $\mathbf{u}^\top \mathbf{G} \mathbf{p} = \mathbf{p}^\top \mathbf{D} \mathbf{u}$	Machine ε	✓
BC1	bc_D	Global mass balance $\sum \mathbf{bc}_D = 0$	Exact	✓
BC2	bc_L	Top-wall BC value $= 2U^*/\Delta y^2$	Exact	✓
L1	L	Laplacian of quadratic field $= 2$	Exact interior	✓
L2	L	Laplacian of linear field $= 0$	Machine ε	✓
L3	L	Self-adjoint: $\mathbf{u}^\top \mathbf{L} \mathbf{v} = \mathbf{v}^\top \mathbf{L} \mathbf{u}$	Machine ε	✓
A1	A	Uniform flow gives zero advection	Machine ε	✓
A2	A	Energy: $\mathbf{u}^\top \mathbf{A}(\mathbf{u}) \approx 0$	Small ($O(\Delta x^2)$)	✓
CG1	CG	Solves Poisson problem to tolerance	$< \text{tol}$	✓
FULL	T	$\mathbf{p}^\top \mathbf{T} \mathbf{p} > 0$ (SPD)	Positive	✓

V. RESULTS

V.A. Grid and Temporal Convergence

All simulations were run on a 65×65 uniform grid with $\text{CFL} = 0.1$, giving a time step $\Delta t = \text{CFL} \Delta x \approx 1.54 \times 10^{-3}$. Time integration was advanced to $t \approx 38.5$ (25,001 steps), sufficient for all three Reynolds numbers to reach a steady state. The fractional-step approximation $A^{-1} \approx \Delta t I$ in the block-LU splitting introduces a splitting error of $O(\Delta t)$, so overall temporal convergence is first-order. The staggered-grid spatial operators are second-order accurate (confirmed by unit tests G2, D2, and L1), yielding second-order spatial convergence. The 65×65 grid resolves the primary vortex and secondary corner structures documented in Ghia et al., and the resulting centerline profiles show good agreement with the reference data at all three Reynolds numbers tested.

V.B. Validation Against Ghia et al.

Figure 3 compares the computed centerline velocity profiles against the reference data of Ghia, Ghia, and Shin for $Re = \{100, 400, 1000\}$. Panel (a) shows u along the vertical centreline $x = 0.5$ and panel (b) shows v along the horizontal centreline $y = 0.5$. The present results match the benchmark closely at $Re = 100$ and $Re = 400$. At $Re = 1000$, small discrepancies near the peak values are consistent with the expected limitation of the uniform 65×65 grid in resolving the thinning near-wall boundary layers.

V.C. Streamline Topology

The rightmost panels of Figure 3 show streamline patterns for $Re = 100, 400$, and 1000 . At $Re = 100$, a single primary recirculation vortex dominates the cavity with its center located near the geometric center of the domain. As Re increases to 400 , the primary vortex shifts toward the cavity center and secondary vortices begin to form in the lower corners. At $Re = 1000$, the primary vortex occupies the central region and the secondary corner vortices have grown substantially, consistent with the

flow topology described by Ghia et al.

V.D. Vorticity Contours

The u and v velocity contours in Figure 2 qualitatively reflect the vorticity distribution at each Reynolds number. At $Re = 100$, the velocity gradients are moderate and distributed across the cavity. As Re increases to 1000 , the gradients concentrate near the walls, particularly along the driven lid and in the lower corners, indicating thin vorticity-bearing boundary layers. This increasing gradient concentration at higher Reynolds numbers is a direct consequence of the reduced viscous diffusion relative to inertia, and represents the primary resolution challenge for uniform-grid solvers at high Re .

V.E. Additional Exploration

The right column of Figure 2 shows pressure field contours for each Reynolds number. The pressure distribution exhibits a high-pressure stagnation region near the upper-right corner, where the lid velocity meets the stationary right wall, and a low-pressure region near the upper-left corner. This pressure gradient is required to sustain the primary recirculation vortex through centrifugal balance. As Re increases, the pressure extrema intensify and the isobars become more tightly packed near the corner regions, reflecting the stronger inertial forces and sharper velocity gradients that develop at higher Reynolds numbers. The pressure is defined only up to a constant and is pinned to zero at the lower-left cell to provide a reference.

VI. DISCUSSION

The fractional-step method couples temporal and spatial errors in two distinct ways. In time, the block-LU decomposition approximates $A^{-1} \approx \Delta t I$, which introduces a splitting error of $O(\Delta t)$; the overall time-marching scheme therefore converges at first order regardless of the Adams–Bashforth and Crank–Nicolson discretizations being second order individually. In space,

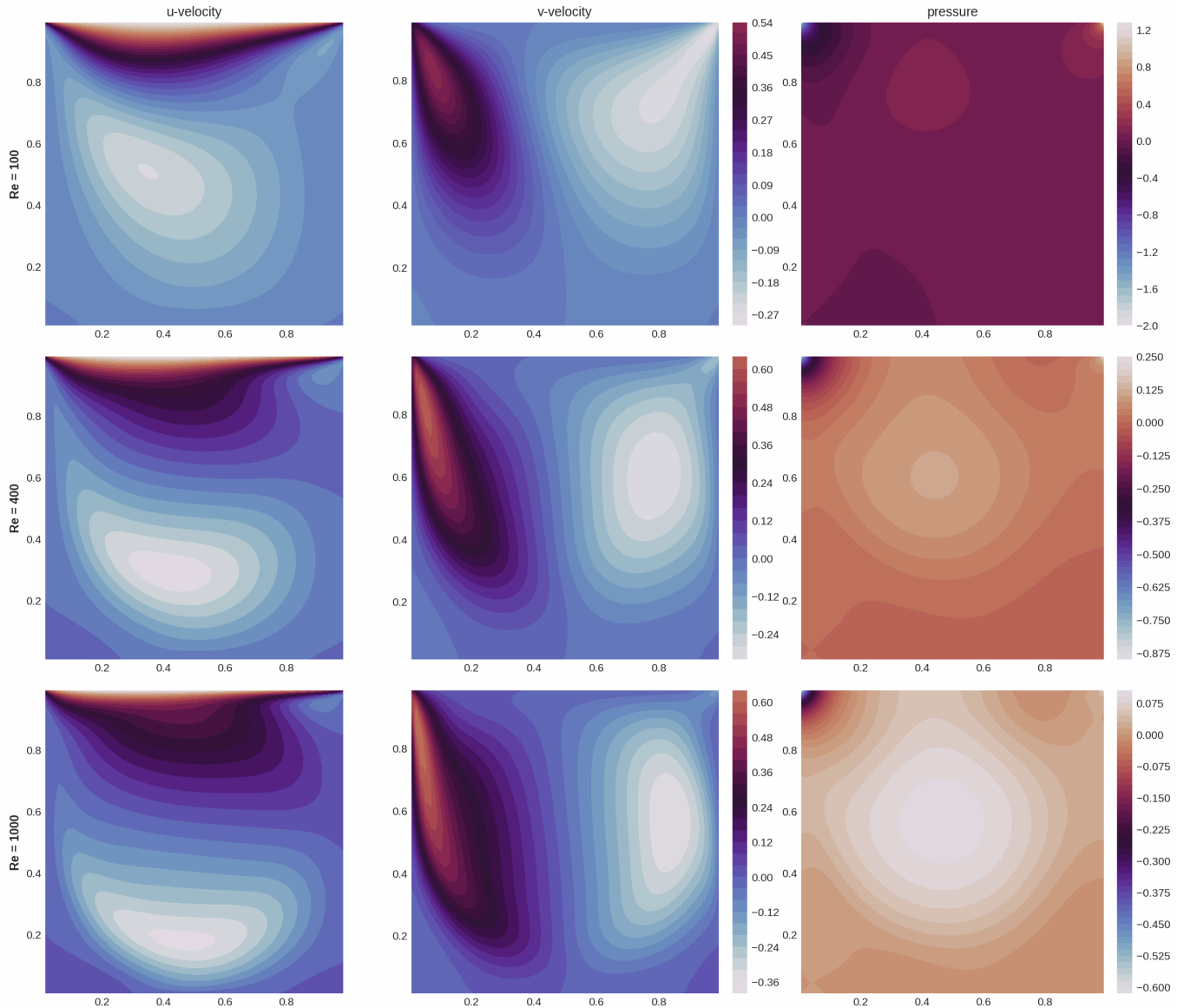


FIG. 2 u and v velocity and pressure contours for $Re = 100, 400, 1000$.

all operators on the staggered grid are second-order accurate: central differences for $\mathbf{L}$ and $\mathbf{N}$, and consistent forward/backward differences for $\mathbf{G}$ and $\mathbf{D}$ whose composition $\mathbf{DG}$ recovers the scalar Laplacian to second order.

Agreement with the Ghia et al. benchmark is excellent at $Re = 100$ and $Re = 400$. At $Re = 1000$, small discrepancies in the peak centerline velocities are expected because the uniform 65×65 grid underresolves the thin boundary layers that form near the walls. A non-uniform grid refined toward the walls, or a finer uniform grid, would reduce these errors.

Physically, the results confirm the well-known vortex topology of the lid-driven cavity: a primary recirculation vortex that migrates toward the cavity center as Re increases, accompanied by growing secondary vortices in the lower corners. The pressure distribution is consistent with the centrifugal balance required to sustain the pri-

mary vortex, with a high-pressure stagnation region at the upper-right corner and a low-pressure region at the upper-left corner.

VII. CONCLUSION

A fractional-step projection solver for the 2-D incompressible Navier–Stokes equations was implemented on a staggered MAC grid and validated against the benchmark lid-driven cavity results of Ghia et al.. The method couples second-order Adams–Bashforth advection and Crank–Nicolson diffusion with a conjugate gradient pressure Poisson solve that enforces the divergence-free constraint at each time step. Simulations on a 65×65 grid with $CFL = 0.1$ reproduced the reference centerline velocity profiles for $Re = 100, 400$, and 1000 with close agreement, and the computed streamline and pressure

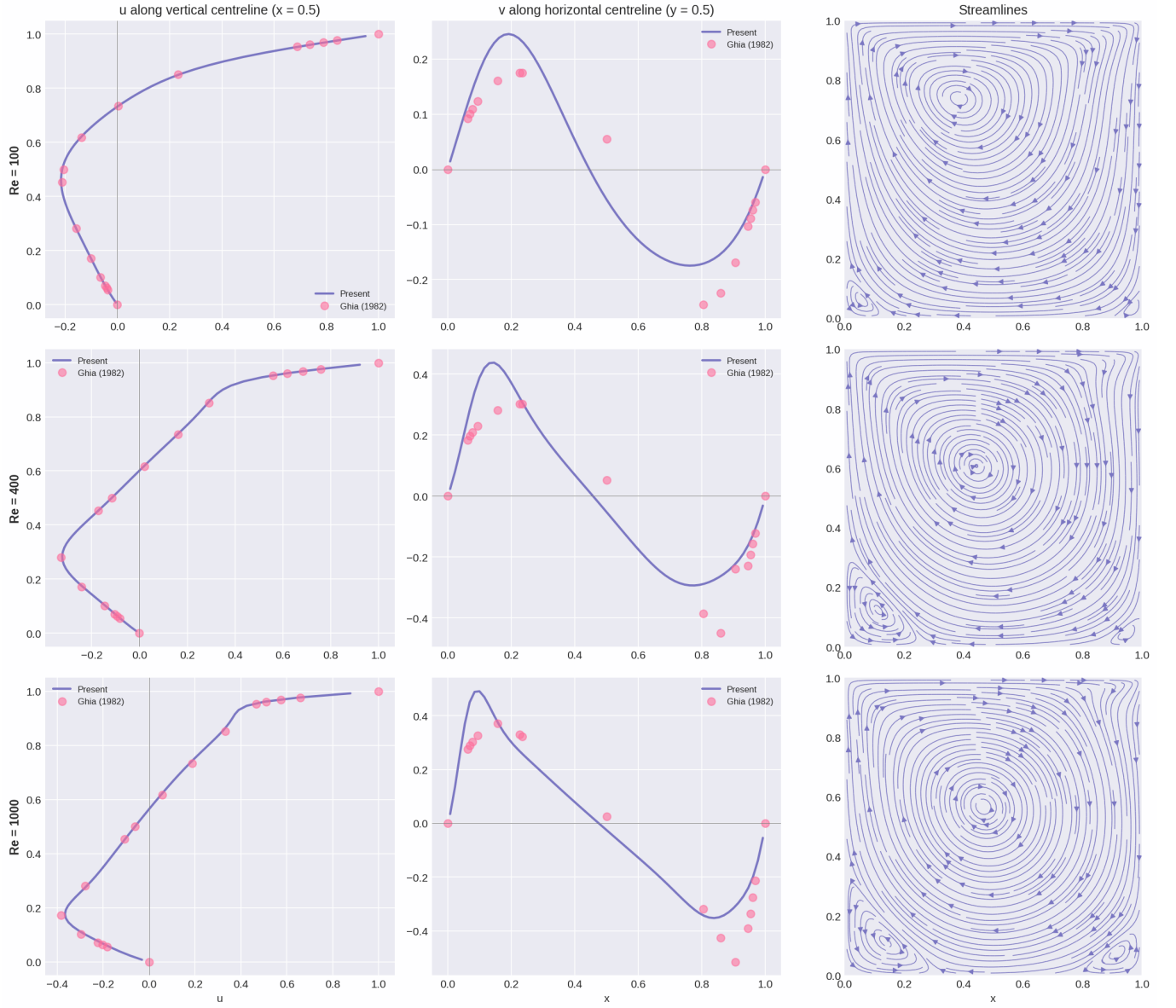


FIG. 3 Top row: (a) horizontal velocity u along the vertical centerline $x = 0.5$ and (b) vertical velocity v along the horizontal centerline $y = 0.5$ for $Re = 100$, 400 , and 1000 . Solid lines: present simulation; symbols: Ghia et al. Right column: streamline contours for each Reynolds number.

fields show the expected flow topology. The spatial operators are second-order accurate on the staggered grid, while the fractional-step splitting limits temporal accuracy to first order. At $Re = 1000$, small deviations from

the reference data highlight the limitation of a uniform grid in resolving thin near-wall boundary layers; these could be addressed with a wall-refined non-uniform grid or a higher-order temporal splitting scheme.